ЛАБОРАТОРНАЯ РАБОТА · ОТЧЕТ О ВЫПОЛНЕНИИ

Git: от локального репозитория до Pull Request

Репозиторий: https://github.com/semyoonov/intro_to_the_profession_lab1

Задача 1. Создать репозиторий

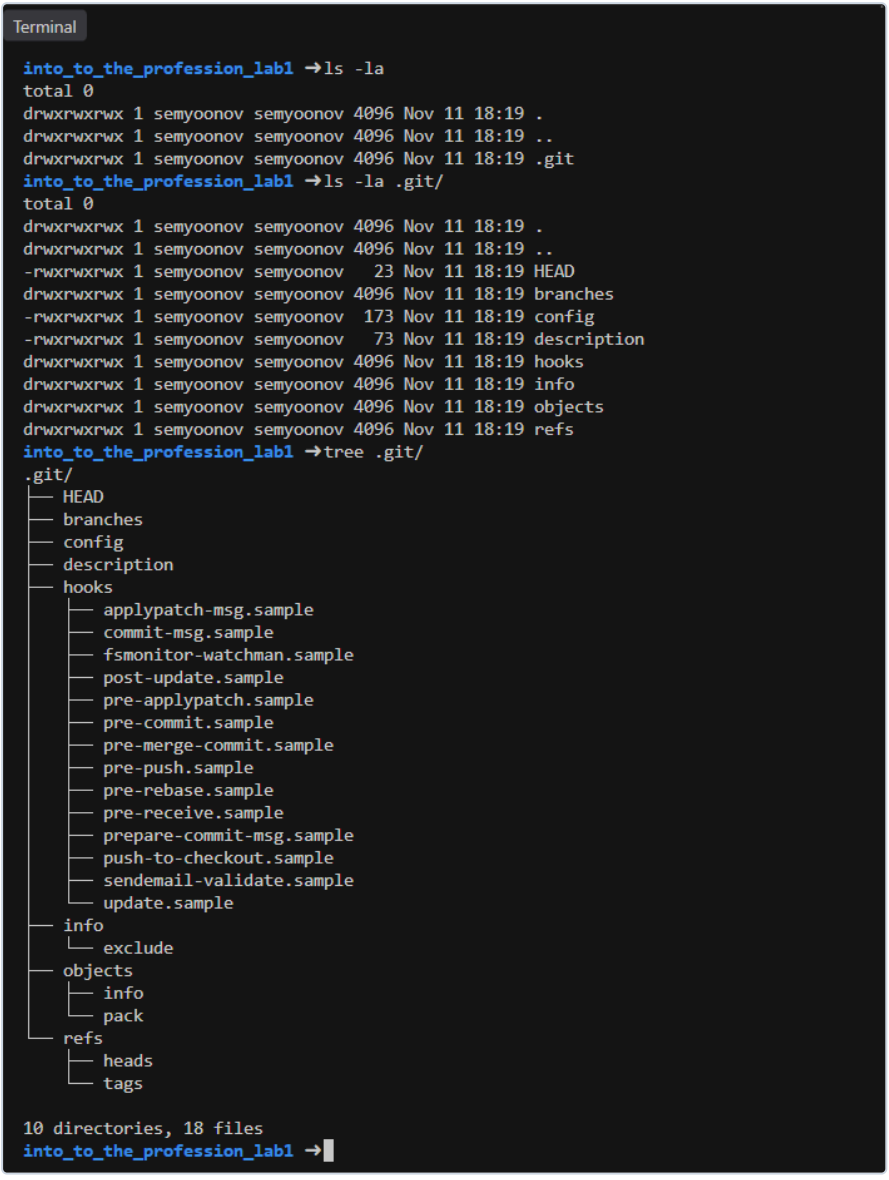


Рис. 1. Инициализация локального репозитория и структура служебной директории .git (HEAD, refs/heads, objects).

Задача 2. Разобраться с состояниями файлов

> Рабочая директория – файлы в их последних состояниях. Index – снапшот, который мы сделали для гита.
>
> Если файл уже затрекан, то `.gitignore` не поможет, нужно сделать `git rm --cached`.

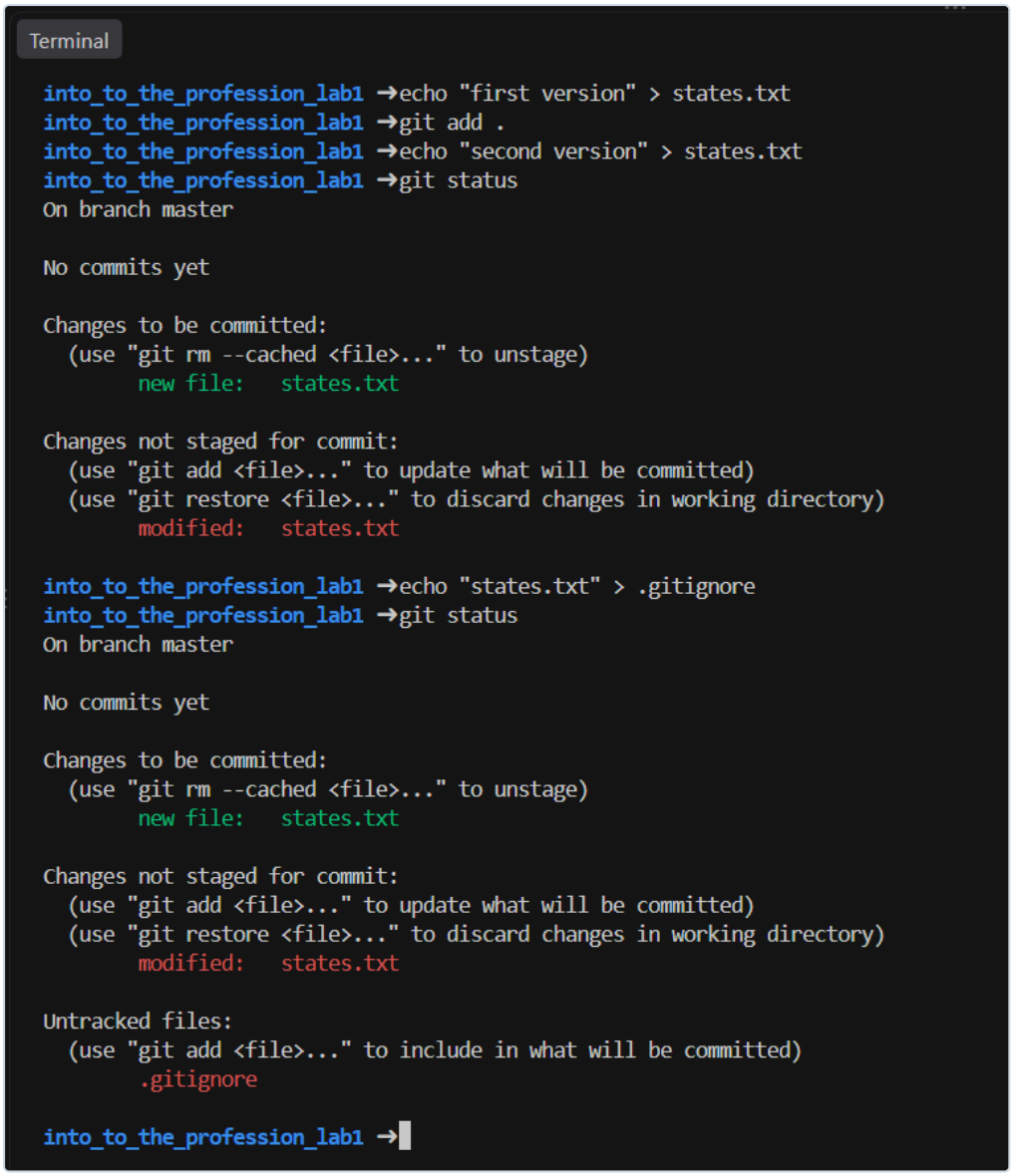


Рис. 2. Одновременное наличие staged и unstaged изменений для states.txt и проверка поведения .gitignore.

Задача 3. Заглянуть внутрь Git

```
into_to_the_profession_lab1 →git commit -m "first commit"
[master (root-commit) 2df9d5a] first commit
 1 file changed, 1 insertion(+)
 create mode 100644 states.txt
into_to_the_profession_lab1 →echo "this file is for my new commit" > state_for_the_second_commit.txt
into_to_the_profession_lab1 →git add state_for_the_second_commit.txt
into_to_the_profession_lab1 →git commit -m "second commit"
[master 277d753] second commit
 1 file changed, 1 insertion(+)
 create mode 100644 state_for_the_second_commit.txt
into_to_the_profession_lab1 →git log --oneline

277d753 (HEAD -> master) second commit
2df9d5a first commit
into_to_the_profession_lab1 →git cat-file -p HEAD
tree 922c634ec9909f3b5ad0889fa79512a7f7424544
parent 2df9d5ab98f6020d49536bcde69abc3133dddec2
author semyoonov <timofei.semyonov@gmail.com> 1794411069 +0300
committer semyoonov <timofei.semyonov@gmail.com> 1794411069 +0300

second commit
into_to_the_profession_lab1 →git cat-file -p HEAD^{tree}
100644 blob af2e75dc40b28de4f93244bf34266da324db34e4    state_for_the_second_commit.txt
100644 blob 22de8d69c9026be2a49f540fda12f3e755a33e6c    states.txt
```

Рис. 3.1. Исследование объектов commit, tree и blob с помощью git cat-file.

```
into_to_the_profession_lab1 →git hash-object states.txt
ad7ac37bb280ccd34b350a59ba440614d9106e41
into_to_the_profession_lab1 →echo "few changes" > states.txt
into_to_the_profession_lab1 →git hash-object states.txt
5a3d5e33a254348fa89ae6caa2fdd861b1ac4554
into_to_the_profession_lab1 →
```

Рис. 3.2. Изменение SHA-хеша объекта states.txt после внесения изменений в файл.

```
Terminal

into_to_the_profession_lab1 →git rev-parse HEAD
277d753ef4afcb35515b8b42debba7610849fae5
into_to_the_profession_lab1 →git cat-file -p HEAD

tree 922c634ec9909f3b5ad0889fa79512a7f7424544
parent 2df9d5ab98f6020d49536bcde69abc3133dddec2
author semyoonov <timofei.semyonov@gmail.com> 1794411069 +0300
committer semyoonov <timofei.semyonov@gmail.com> 1794411069 +0300

second commit
into_to_the_profession_lab1 →git ls-tree -r HEAD
100644 blob af2e75dc40b28de4f93244bf34266da324db34e4    state_for_the_second_commit.txt
100644 blob 22de8d69c9026be2a49f540fda12f3e755a33e6c    states.txt
into_to_the_profession_lab1 →
```

Рис. 3.3. Подтверждение SHA-идентификаторов коммита, дерева и blob-объектов.

Задача 4. Поработать с историей и ветками

> Merge создает отдельный коммит, squash нет.

```
into_to_the_profession_lab1 ➜git log --graph --oneline --all

* d8699f2 (HEAD -> main) feat: squash commit
| * c383873 (feature-squash) second commit to test squash
| * d9ffa0d first commit to test squash
|/
*   8b350de Merge branch 'feature-merge'
|\
| * c9b012c (feature-merge) feat: merge branch
* | c5c15cd feat: main branch work
|/
* fac0ba9 feat: add line 5
* 7f894ca feat: add line 4
* 134770d feat: add line 3
* 6da3029 feat: add line 2
* 2414ec3 feat: add line 1
* 277d753 second commit
* 2df9d5a first commit
into_to_the_profession_lab1 ➜
```

Рис. 4. Граф истории коммитов (git log --graph --oneline --all) после обычного слияния и squash.

Задача 5. Связать локальный Git с GitHub

- Репозиторий: https://github.com/semyoonov/intro_to_the_profession_lab1

```
To https://github.com/semyoonov/into_to_the_profession_lab1.git
   d8699f2..b3c03b2  main -> main
into_to_the_profession_lab1 ➜cd ..
project ➜ls
clone_intro_to_the_profession  into_to_the_profession_lab1
project ➜cd clone_intro_to_the_profession
clone_intro_to_the_profession ➜git fetch origin
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 3 (delta 1), reused 3 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 278 bytes | 2.00 KiB/s, done.
From https://github.com/semyoonov/into_to_the_profession_lab1
   d8699f2..b3c03b2  main       -> origin/main
clone_intro_to_the_profession ➜git log --oneline origin/main
b3c03b2 (origin/main, origin/HEAD) feat: commit for fetch
d8699f2 (HEAD -> main) feat: squash commit
8b350de Merge branch 'feature-merge'
c5c15cd feat: main branch work
c9b012c feat: merge branch
fac0ba9 feat: add line 5
7f894ca feat: add line 4
134770d feat: add line 3
6da3029 feat: add line 2
2414ec3 feat: add line 1
277d753 second commit
2df9d5a first commit
clone_intro_to_the_profession ➜git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)

nothing to commit, working tree clean
clone_intro_to_the_profession ➜
```

Рис. 5. Состояние второй локальной копии после git fetch origin до синхронизации с удалённой веткой.

Задача 6. Подготовить GitHub workflow

- Pull Request Template: https://github.com/semyoonov/intro_to_the_profession_lab1/blob/main/.github/pull_request_template.md
- GitHub Actions Workflow: https://github.com/semyoonov/intro_to_the_profession_lab1/blob/main/.github/workflows/ci.yml

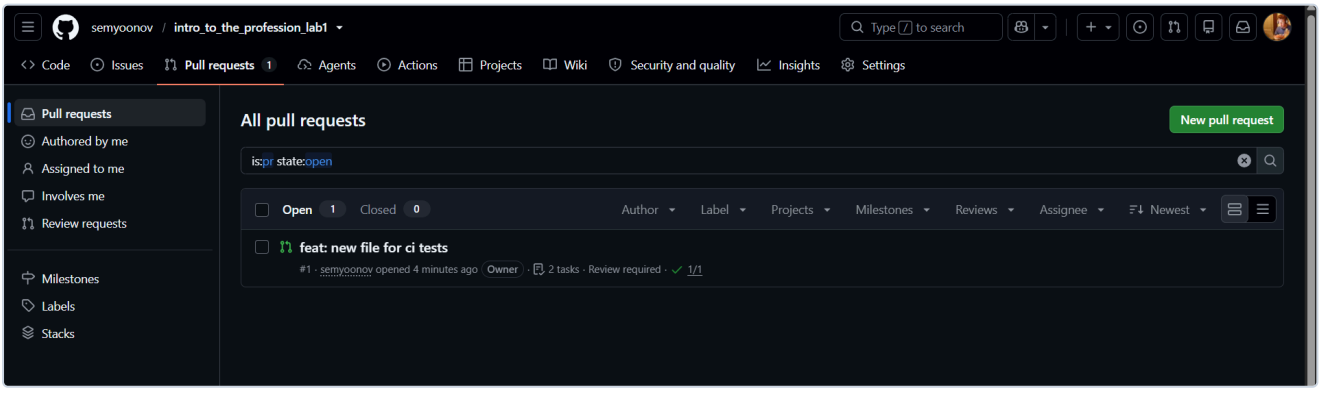


Рис. 6. Pull Request с подключённым шаблоном задач и обязательной проверкой CI.

Задача 7. Создать PR и намеренно получить конфликт

- Issue #2: https://github.com/semyoonov/intro_to_the_profession_lab1/issues/2
- Pull Request #3: https://github.com/semyoonov/intro_to_the_profession_lab1/pull/3

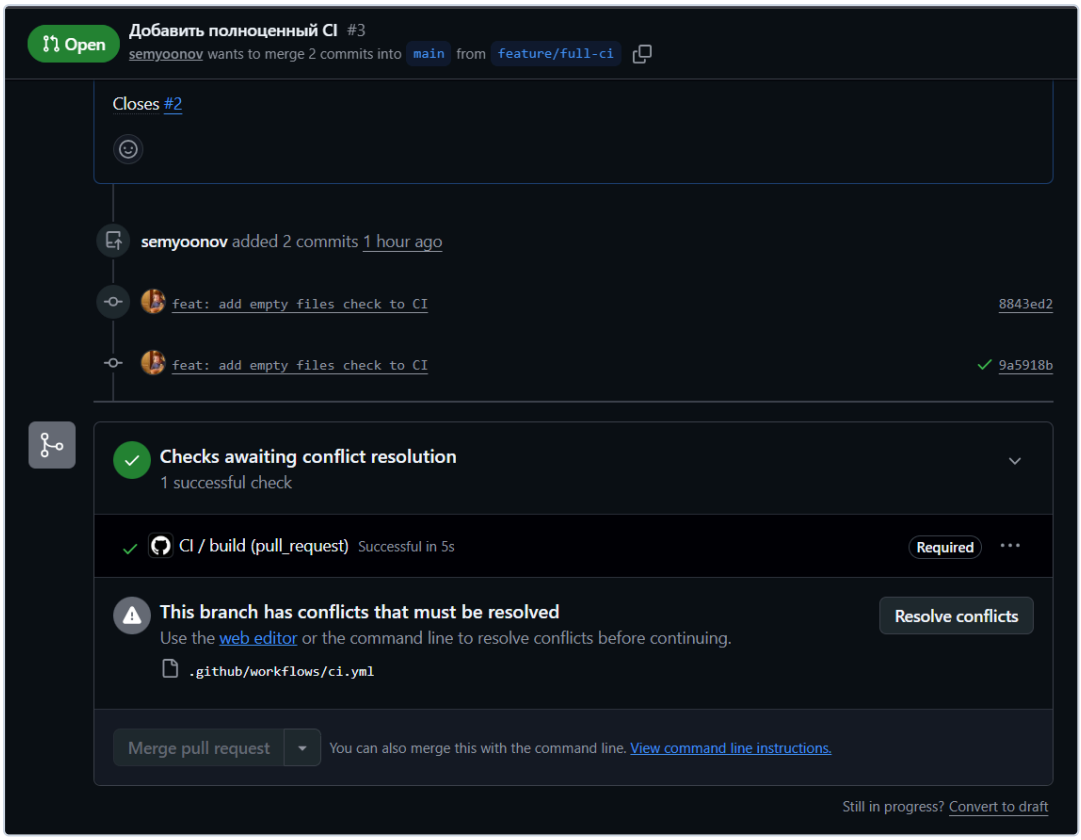


Рис. 7. Pull Request #3 в состоянии конфликта с веткой main в файле .github/workflows/ci.yml.

Задача 8. Пройти настоящее Code Review

- Pull Request #3: https://github.com/semyoonov/intro_to_the_profession_lab1/pull/3

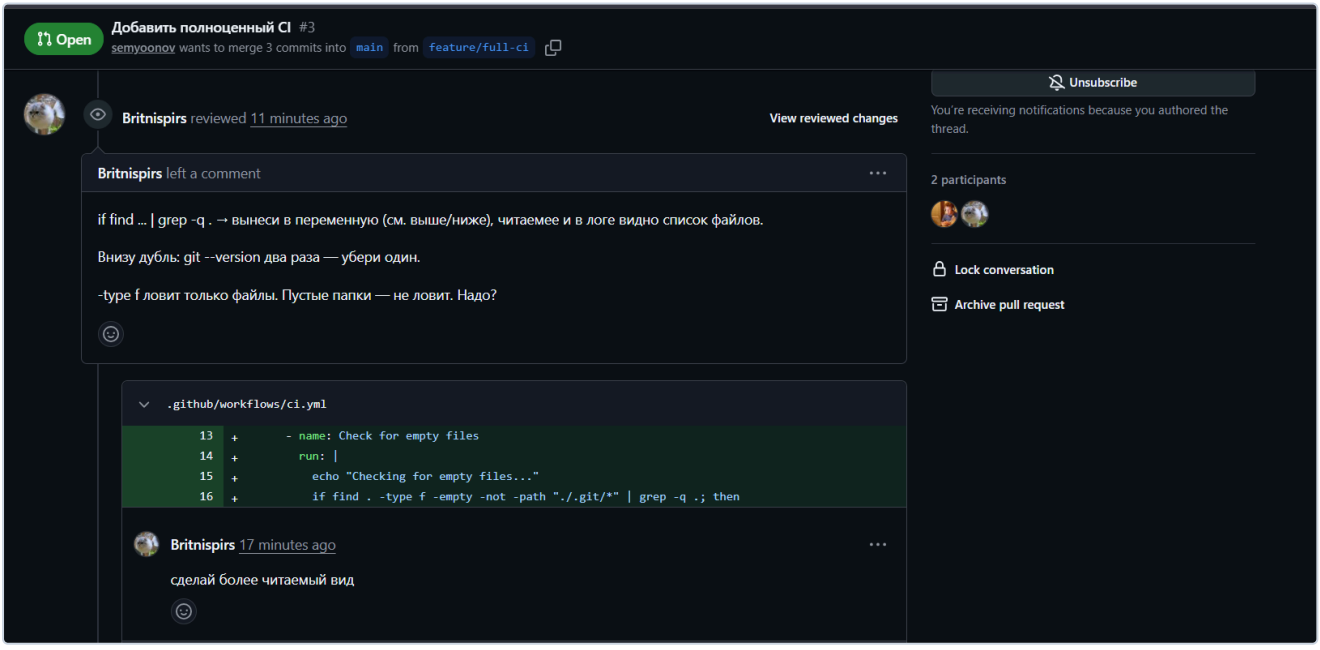


Рис. 8.1. Замечания ревьюера по коду workflow в Pull Request #3.

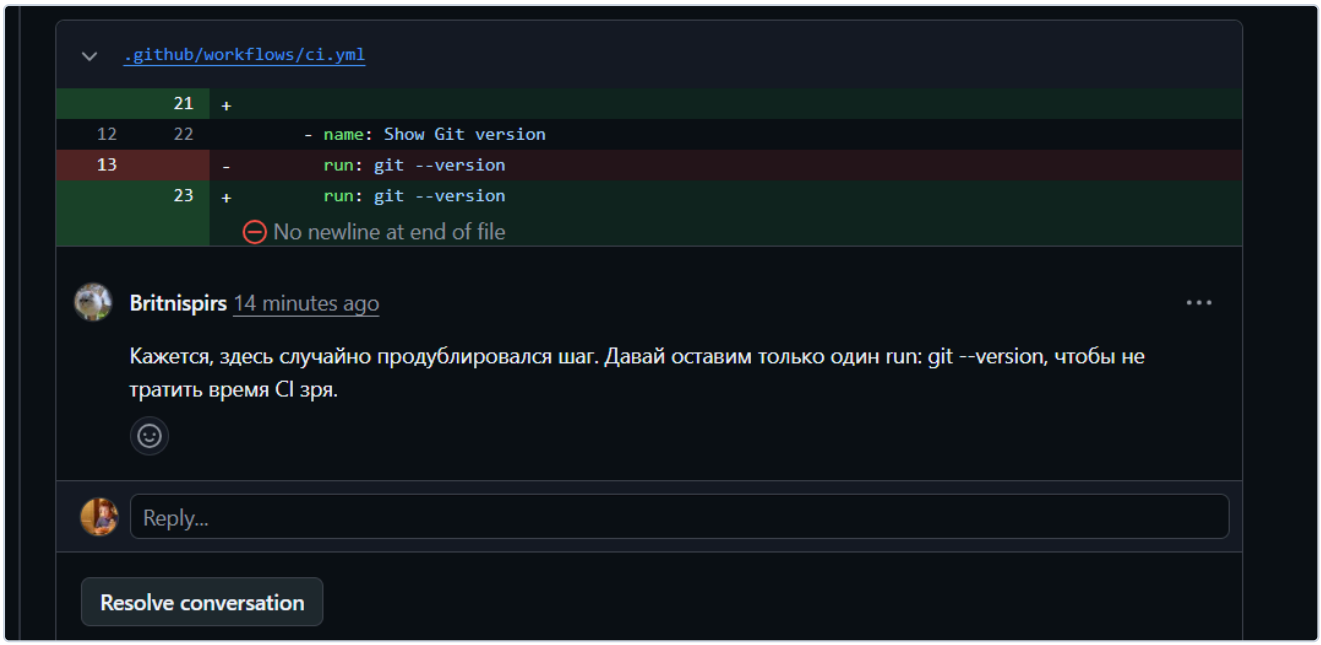


Рис. 8.2. Второе inline-замечание ревьюера по дублированию шага в ci.yml.

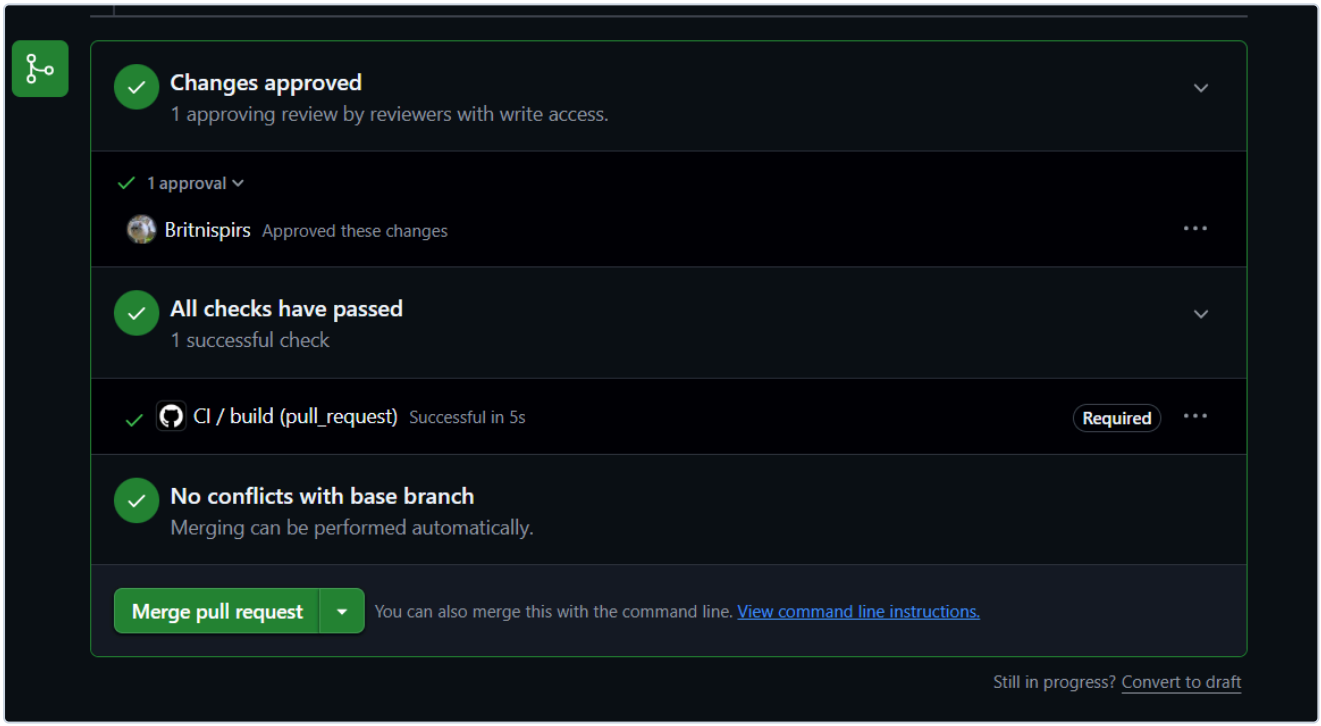


Рис. 8.3. Подтверждение изменений (Changes approved) и зелёный статус CI.

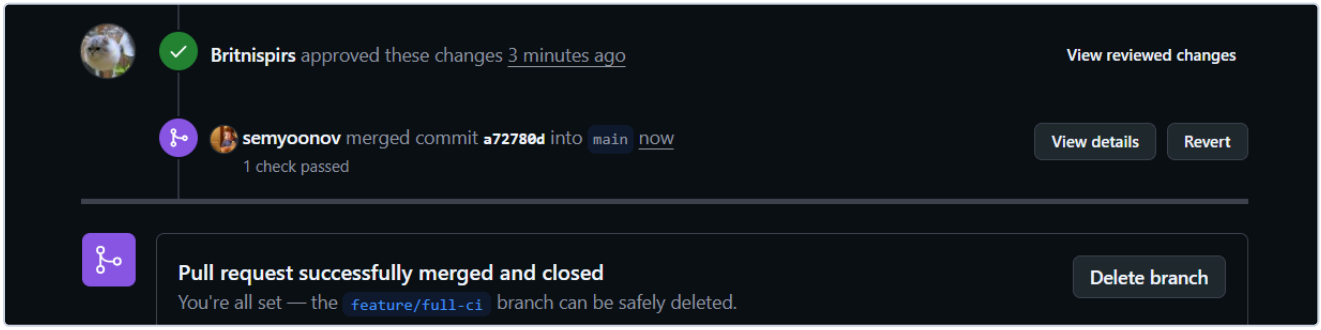


Рис. 8.4. Успешное слияние Pull Request #3 в ветку main.

Задача 9. Сломать CI секретом

- Pull Request #5: https://github.com/semyoonov/intro_to_the_profession_lab1/pull/5
- Failed Actions Run: https://github.com/semyoonov/intro_to_the_profession_lab1/actions/runs/43101479509

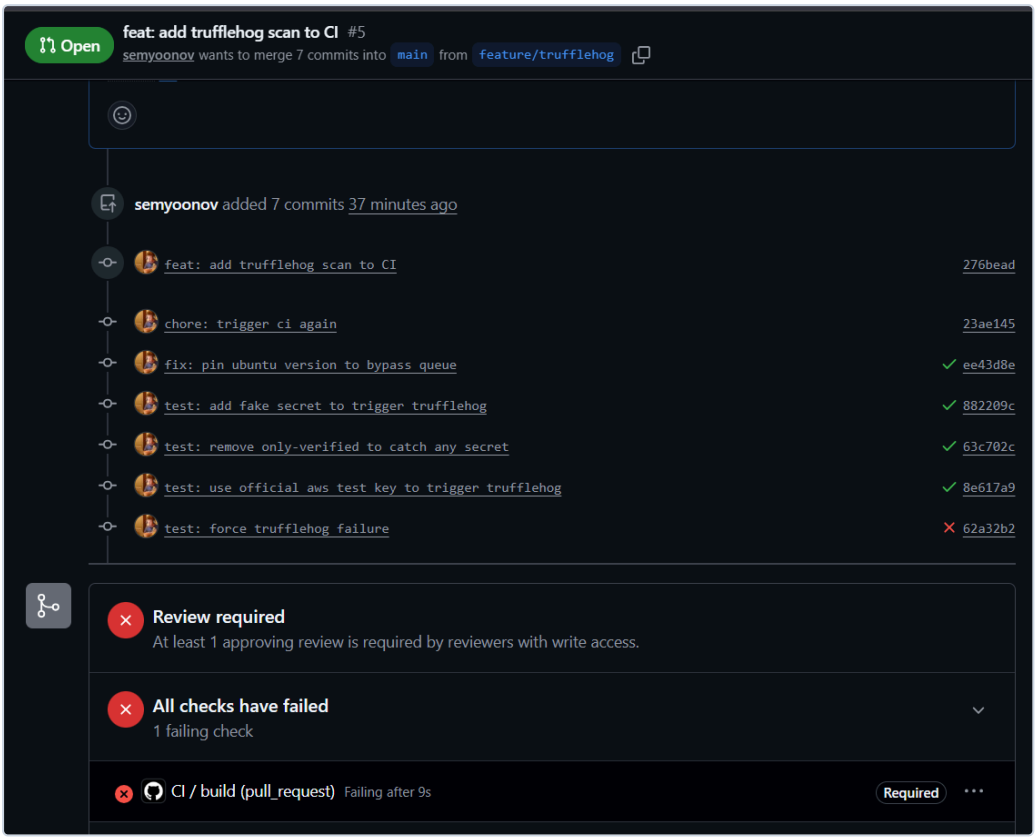


Рис. 9. Ошибка CI в Pull Request #5 после добавления тестового ключа для проверки TruffleHog.

Задача 10. Починить CI и завершить workflow

- Pull Request #5: https://github.com/semyoonov/intro_to_the_profession_lab1/pull/5
- Successful Actions Run: https://github.com/semyoonov/intro_to_the_profession_lab1/actions/runs/43376043438

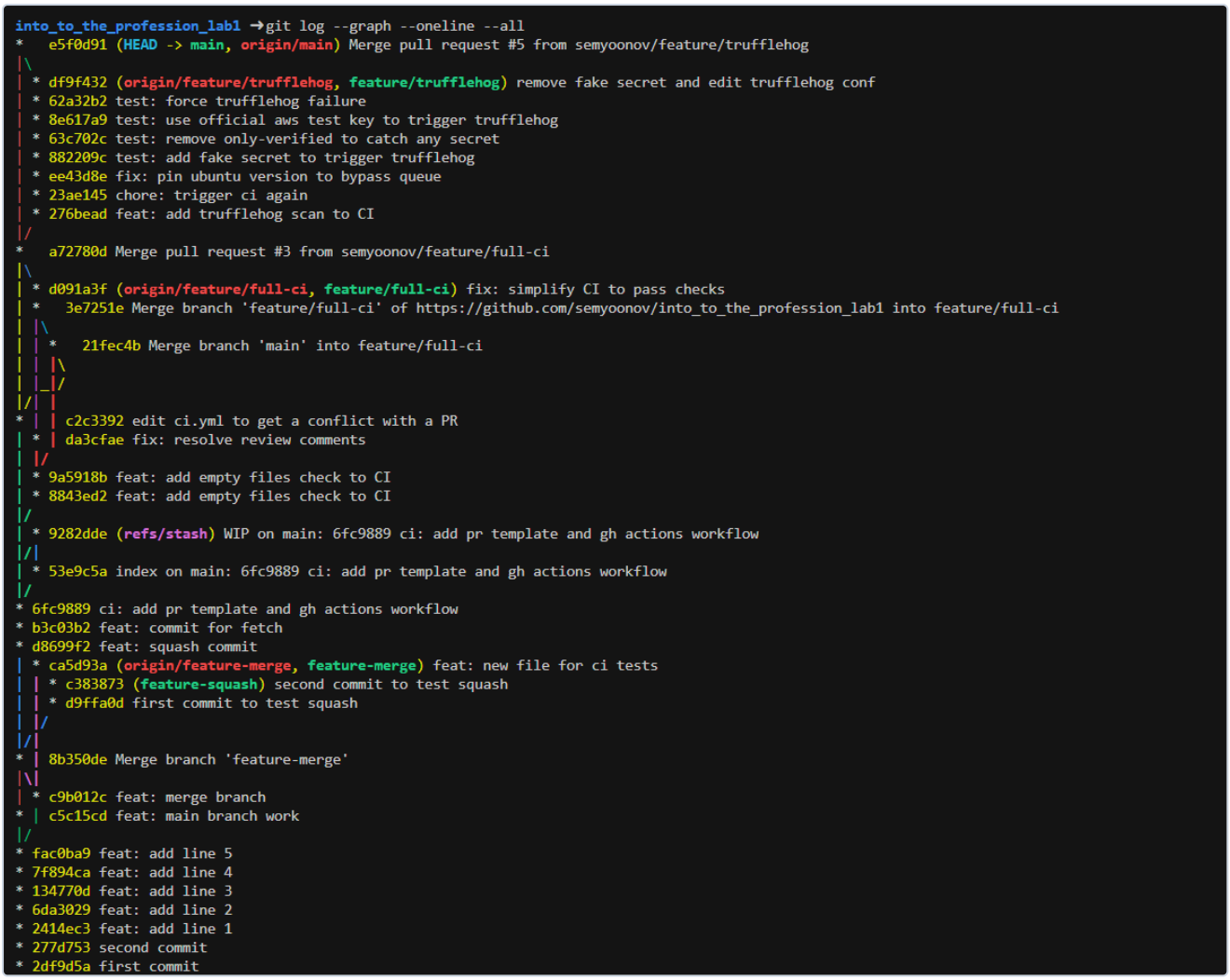


Рис. 10. Финальный граф истории репозитория после исправления CI и слияния PR #5 в main.